

RESEARCH

& INNOVATION UPDATES

High-power wireless EV charging made possible



ORNL's Omer Onar, left, power electronics research lead, discussing the 270kW wireless charging system's lightweight polyphase electromagnetic coupling coil, pictured in foreground, with Lee Slezak (centre), DOE's Vehicle Technologies Office technology manager for grid and charging infrastructures (Credit: Carlos Jones/ORNL, US Dept. of Energy)

Oak Ridge National Laboratory (ORNL) and Volkswagen Group of America demonstrated a 270kW wireless power transfer to a Porsche Taycan, a light-duty electric vehicle. ORNL's poly-phase wireless charging system overcame space, weight, and volume challenges typical in large systems. Current solutions for light-duty vehicles target up to 11kW, achieving 92% efficiency. ORNL's system used polyphase electromagnetic coils, 48.26cm wide, to maximise power density, similar to consumer device charging. The system, achieving a 50% charge increase in 10 minutes with over 95% efficiency, successfully transferred power across a 12.06cm gap, ensuring safety and reliability with protective measures.

Reusable, environment-friendly UV bottle for potable water

Het Dave, an MDes student at Anant National University, under the guidance of Professor Parth Shah, has designed 'IRA—A UV water bottle' after extensive primary and secondary research into existing technologies. They developed a compact solution using UV-C light, which effectively disrupts the DNA of harmful microorganisms to ensure water safety. The resulting product is a glass bottle equipped with a UV-C light circuit, crafted into a user-friendly design through rigorous development and testing. This bottle not only purifies water but also reduces reliance on single-use plastics by providing a reusable, environmentally responsible alternative. The design process included meticulous planning of the electronic components, such as circuit designs that accommodate UV-C light, rechargeable batteries, and electronic controls - all within the confined space of the bottle.

IRA-A UV water bottle—Change the way you hydrate
(<https://www.electronicsforu.com>)

